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NETWORKING SUBSYSTEM/SWITCH DEVICE CONNECTION CONFIGURATION SYSTEM

Abstract

A networking subsystem/switch device connection configuration system includes a switch device having a switch physical port, and an endpoint device including a networking subsystem having a plurality of networking physical ports. The networking subsystem detects connection of respective cable connectors to each of the plurality of networking physical ports. If the networking subsystem determines that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, it configures a single networking logical port to transmit data between each of the plurality of networking physical ports and the switch physical port.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates generally to information handling systems, and more particularly to configuring a connection between a switch device and a networking subsystem in an information handling system.

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

[0003] Information handling systems such as, for example, endpoint devices provided by server devices, storage systems, and/or other endpoint devices known in the art, are often connected to switch devices via their networking subsystems (e.g., Network Interface Controller (NIC) devices, network adapter devices, and/or other networking subsystems known in the art) in order to transmit data via a network. As will be appreciated by one of skill in the art, while switch devices have traditionally been upgraded to newer generations prior to networking subsystems, networking subsystems have begun being upgraded to newer generations prior to switch devices due to, for example, the relatively lower cost of networking subsystem as compared to switch devices.

[0004] To provide a specific example, relatively older generation switch devices may currently utilize Quad Form-Factor Pluggable Double Density (QSFP-DD) form factor physical ports that include eight 56 Gigabits per second (Gbps) lanes (e.g., to provide a 400G effective data transmission rate capability), while relatively new generation networking subsystems may currently utilize Quad Form-Factor Pluggable 112 Gbps (QSFP112) form factor physical ports that include four 112 Gbps lanes (e.g., to provide a 400G effective data transmission rate capability). However, such relatively older generation switch devices are not conventionally directly connected to such relatively newer generation networking subsystems by passive cables such as conventional 400G Direct Attach Cables (DACs) due to the different number of lanes and speeds of the QSFP-DD form-factor physical ports and the QSFP112 form-factor physical ports.

[0005] Rather, conventional solutions to connect such relatively older generation switch devices to such relatively newer generation networking subsystems include the connection of respective optical modules on each of the QSFP-DD form-factor physical port(s) and the QSFP112 form-factor physical port(s), followed by the connection of the common optical interfaces on those optical modules using an optical cable. However, such optical modules are relatively expensive and consume relatively high amounts of power as compared to relatively low-cost passive cables such as the conventional DACs discussed above (e.g., DACs are typically half the cost of the optical coupling system described above, and consume no power as compared to the approximately 12 watts consumed by each of the optical modules described above).

[0006] Accordingly, it would be desirable to provide and configure a connection between a networking subsystem and a switch device in a manner that addresses the issues discussed above.

SUMMARY

[0007] According to one embodiment, an Information Handling System (IHS) includes a processing system; and a memory system that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a port configuration engine that is configured to: detect connection of respective cable connectors to each of a plurality of networking physical ports that are coupled to the processing system; determine that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable; and configure, in response to determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, a single networking logical port to transmit data between each of the plurality of networking physical ports and a switch physical port on a switch device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a schematic view illustrating an embodiment of an Information Handling System (IHS).

[0009] FIG. **2** is a schematic view illustrating an embodiment of a networked system that may provide the networking subsystem/switch device connection configuration system of the present disclosure.

[0010] FIG. **3** is a flow chart illustrating an embodiment of a method for configuring a connection between a networking subsystem and a switch device.

[0011] FIG. **4**A is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0012] FIG. **4**B is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0013] FIG. **4**C is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0014] FIG. **5**A is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0015] FIG. **5**B is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0016] FIG. **5**C is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0017] FIG. **6**A is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

[0018] FIG. **6**B is a schematic view illustrating an embodiment of the networked system of FIG. **2** operating during the method of FIG. **3**.

DETAILED DESCRIPTION

[0019] For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0020] In one embodiment, IHS **100**, FIG. **1**, includes a processor **102**, which is connected to a bus **104**. Bus **104** serves as a connection between processor **102** and other components of IHS **100**. An input device **106** is coupled to processor **102** to provide input to processor **102**. Examples of input devices may include keyboards, touchscreens, pointing devices such as mouses, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device **108**, which is coupled to processor **102**. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS **100** further includes a display **110**, which is coupled to processor **102** by a video controller **112**. A system memory **114** is coupled to processor **102** to provide the processor with fast storage to facilitate execution of computer programs by processor **102**. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis **116** houses some or all of the components of IHS **100**. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor **102** to facilitate interconnection between the components and the processor **102**.

[0021] Referring now to FIG. **2**, an embodiment of a networked system **200** is illustrated that may provide the networking subsystem/switch device connection configuration system of the present disclosure. In the illustrated embodiment, the networked system **200** includes a switch device **202**. In an embodiment, the switch device **202** may be provided by the IHS **100** discussed above with reference to FIG. **1**, and/or may include some or all of the components of the IHS **100**. However, while illustrated and discussed as a switch device **202**, one of skill in the art in possession of the present disclosure will recognize that switch devices provided in the networked system **200** may include any devices that may be configured to operate similarly as the switch device **202** discussed below. In the illustrated embodiment, the switch device **202** includes a chassis **202*a*** that houses the components of the switch device **202**, only some of which are illustrated and described below.

[0022] For example, the chassis **202*a*** may house a networking processing system **204** such as, for example, a Network Processing Unit (NPU) that one of skill in the art in possession of the present disclosure would appreciate is provided in switch devices. Furthermore, in the illustrated embodiment, the chassis **202*a*** may also include a plurality of physical ports **206*a*** and up to **206*b*** (also referred to as "switch physical ports" below) that are coupled to the networking processing system **204** and that are accessible on a surface of the chassis **202*a***. In the specific examples provided below, the switch device **202** is a "first generation" switch device having respective "first generation" Ethernet ports that provide each the physical ports **206*a*-206*b*** and that each include a Quad Form-Factor Pluggable Double Density (QSFP-DD) form-factor with eight lanes that are each configured to transmit data at 56 Gigabits per second (Gbps), which one of skill in the art in possession of the present disclosure will appreciate provides an effective data transmission rate capability of 400G for each of those "first generation" Ethernet ports/physical ports **206*a*-206*b***.

[0023] As will be appreciated by one of skill in the art in possession of the present disclosure, the use of "first generation" for the switch devices described herein is provided to indicate the relatively older generation of those switch devices as compared to the networking subsystems described herein, and is not meant to indicate any particular generation of those switch devices. Furthermore, while specific examples of particular characteristics of the switch device **202** and its physical ports **206*a*-206*b*** are described herein, one of skill in the art will appreciate how the teachings of the present disclosure may be utilized with a variety of different generation switch devices having different physical ports with different characteristics than those discussed herein while remaining within the scope of the present disclosure as well.

[0024] In the illustrated embodiment, the networked system **200** also includes an endpoint device **208**. In an embodiment, the endpoint device **208** may be provided by the IHS **100** discussed above with reference to FIG. **1**, and/or may include some or all of the components of the IHS **100**, and may be provided by server devices, storage systems, and/or any other endpoint devices that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, while particular endpoints devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will recognize that endpoint devices provided in the networked system **200** may include any devices that may be configured to operate similarly as the endpoint device **208** discussed below. In the illustrated embodiment, the endpoint device **208** includes a chassis **208***a* that houses the components of the endpoint device **208**, only some of which are illustrated and described below.

[0025] For example, the chassis **208***a* may house a networking subsystem **210** such as, for example, a Network Interface Controller (NIC), a networking adapter device, and/or any other networking subsystems that one of skill in the art in possession of the present disclosure would recognize may be used to connect the endpoint device **208** to switch devices and/or other network components known in the art. In the illustrated embodiment, the networking subsystem **210** in the chassis **208***a* of the endpoint device **208** may include a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**), and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that includes instructions that, when executed by the processing system, cause the processing system to provide a port configuration engine **212** that is configured to perform the functionality of the port configuration engines, port configuration subsystems, and/or networking subsystems described below.

[0026] As will be appreciated by one of skill in the art in possession of the present disclosure, the processing system in many “second generation” networking subsystems like the networking subsystem **210** described herein may provide flexibility and backward compatibility be being provided by 1×400G NIC chip that includes eight 112G SERializer/DESerializer (SERDES) lanes, which one of skill in the art in possession of the present disclosure will appreciate may be used to support the 400G effective data transmission rate embodiments discussed below by providing either 56 Gbps or 112 Gbps per lane. Thus, in one specific example, each of the eight lanes the 1×400G NIC chip may be directly connected to each of the respective eight 56 Gbps lanes on a 1×400G QSFP-DD port. In another specific example, a first set of four lanes on the 1×400G NIC chip may be directly connected to each of the respective four 56 Gbps lanes on a first 1×200G QSFP56 port, and a second set of four lanes on that 1×400G NIC chip may be directly connected to each of the respective four 56 Gbps lanes on a second 1×200G QSFP56 port. In yet another specific example, a first set of four lanes on the 1×400G NIC chip may be directly connected to each of the respective four 112 Gbps lanes on a first 1×400G QSFP112 port, and a second set of four lanes on that 1×400G NIC chip may be directly connected to each of the respective four 112 Gbps lanes on a second 1×400G QSFP112 port, with the second set of four lanes on the 1×400G NIC chip utilized for redundancy.

[0027] In the illustrated embodiment, the networking subsystem **210** in the chassis **208***a* of the endpoint device **208** may also include a plurality of physical ports **214***a* and up to **214***b* (also referred to as “networking physical ports” below) that are coupled to the port configuration engine **212** (e.g., via a coupling between that physical port and the processing system) and that are accessible on a surface of the chassis **208***a*. As will be appreciated by one of skill in the art in possession of the present disclosure, the networking subsystem **210** illustrated and described with reference to FIG. **2** is just one example of a networking subsystem provided according to the teachings of the present disclosure, and other embodiments may provide the port configuration engine **212** separately from the networking subsystem **210** (e.g., via a processing system/memory

system combination in the chassis **208***a* that is coupled to the networking subsystem **210**), and/or may provide the networking subsystem **210** in other configurations that will fall within the scope of the present disclosure as well. As such, the networking subsystem **210** is illustrated with dotted lines in FIG. **2**, and not illustrated in the other Figures provided for examples described below, to indicate that the port configuration engine **212** and physical ports **214***a*-**214***b* may be configured to connect with the switch device **202** regardless of how the networking subsystem **210** is provided.
[0028] In the specific examples provided below, the networking subsystem **210** in the chassis **208***a* of the endpoint device **208** is a "second generation" networking subsystem having respective "second generation" Ethernet ports that provide each the physical ports **214***a*-**214***b* and that each include a Quad Form-Factor Pluggable 112 Gbps (QSFP112) form-factor with four lanes that are each configured to transmit data at 112 Gigabits per second (Gbps), which one of skill in the art in possession of the present disclosure will appreciate provides an effective data transmission rate capability of 400G for each of those "second generation" Ethernet ports/physical ports **214***a*-**214***b*. While not illustrated in FIG. **2**, as described below, the port configuration engine **212** may also be coupled to each of the ports **214***a* and **214***b* via a respective side-band Inter-Integrated Circuit (I2C) bus that may be utilized to access a memory system (e.g., the Electronically Erasable Programmable Read-Only Memory (EEPROM) of a cable connector connected to that port.
[0029] Furthermore, while the examples below describe the networking system **210** in the chassis **208***a* of the endpoint device **208** having a pair of networking physical ports **214***a* and **214***b*, one of skill in the art in possession of the present disclosure will appreciate how the teachings of the present disclosure may be expanded to four networking physical ports **214***a*-**214***b* (or more) while remaining within the scope of the present disclosure as well.
[0030] As will be appreciated by one of skill in the art in possession of the present disclosure, the use of "second generation" for the networking subsystems described herein is provided to indicate the relatively newer generation of those networking subsystems as compared to the switch devices described herein, and is not meant to indicate any particular generation of those networking subsystems. Furthermore, while specific examples of particular characteristics of the networking subsystem **210** and its physical ports **214***a*-**214***b* are described herein, one of skill in the art will appreciate how the teachings of the present disclosure may be utilized with a variety of different generation networking subsystems having different physical ports with different characteristics than those discussed herein while remaining within the scope of the present disclosure as well. However, while a specific networked system **200** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that the networked subsystem/switch device connection configuration system of the present disclosure may include a variety of components and/or component configurations for providing conventional networking subsystem/switch device functionality, as well as the networked subsystem/switch device connection configuration functionality discussed below, while remaining within the scope of the present disclosure as well.
[0031] Referring now to FIG. **3**, an embodiment of a method **300** for configuring a connection between a networking subsystem and a switch device is illustrated. As discussed below, the systems and methods of the present disclosure provide for the configuration of a connection between a relatively newer generation networking subsystem via its networking physical ports and a relatively older generation switch device via its switch physical ports that have more lanes and support a lower data transmission rate than those networking physical ports. For example, the networking subsystem/switch device connection configuration system of the present disclosure may include a switch device having a switch physical port, and an endpoint device including a networking subsystem having a plurality of networking physical ports. The networking subsystem detects connection of respective cable connectors to each of the plurality of networking physical ports. If the networking subsystem determines that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, it configures a single

networking logical port to transmit data between each of the plurality of networking physical ports and the switch physical port. As such, older generation switch devices having switch physical ports with more lanes supporting lower data transmission rates may be connected via a passive cable to newer generation networking devices having networking physical ports with fewer lanes supporting higher data transmission rates, with some embodiments auto-provisioning such connection configurations to eliminate the need for manual configurations that would complicate performing such connection configurations at scale.

[0032] As will be appreciated by one of skill in the art in possession of the present disclosure, embodiments of the method **300** provide for the configuration of a connection between a “first generation” switch device and a “second generation” networking subsystem in an endpoint device automatically in response to detecting that connection and without human intervention (e.g., without the need for a network administrator or other user to run a script to configure that connection and/or perform other manual configuration operations known in the art), which provides benefits when connecting networking subsystems to switch devices at scale. However, as also described below, the teachings of the present disclosure may be applied to provide the novel configuration of the connection between a “first generation” switch device and a “second generation” networking subsystem manually as well.

[0033] The method **300** begins at block **302** where a networking subsystem detects connection of respective cable connectors to each of a plurality of networking physical pots on the networking subsystem. As will be appreciated by one of skill in the art in possession of the present disclosure, the physical ports **214***a*-**214***b* on the networking subsystem **210** of the endpoint device **208** may be connected to the switch device **202** (as well as other switch devices as described below) in a variety of manners, and only some of those connections are illustrated and described below. For example, with reference to FIG. **4**A, a network administrator or other user may provide a breakout cable **400** including a plurality of cable connectors **400***a*, **400***b*, and **400***c* to connect the networking subsystem **210** on the endpoint device **208** to the switch device **202**, and one of skill in the art in possession of the present disclosure will appreciate how the breakout cable **400** may be provided by a DAC (e.g., a 400G DAC) or other “passive” cable known in the art.

[0034] In the specific examples provided herein, the cable connector **400***a* on the cable **400** may be provided by a QSFP-DD cable connector that is configured to connect to the QSFP-DD form-factor physical port **206***a*, while the cable connectors **400***b* and **400***c* may each be provided by respective Quad Form-Factor Pluggable 56 Gbps (QSFP56) cable connectors that are each configured to connect to respective one of the QSFP112 form-factor physical ports **214***a* and **214***b*, although one of skill in the art in possession of the present disclosure will appreciate how other types of cable connectors configured to connect to other types of physical ports will fall within the scope of the present disclosure as well. Furthermore, in some embodiments, each of the cable connectors **400***a*, **400***b*, and **400***c* (as well as any other “breakout” cable connectors on the cable **400** that are coupled to the cable connector **400***a* via the cable **400**, if present) may store a common identifier. For example, a respective memory subsystem (e.g., an Electronically Erasable Programmable Read-Only Memory (EEPROM)) in each of the cable connectors **400***a*, **400***b*, and **400***c* may be programmed with, or may otherwise store, a cable serial number of the cable **400**, “far end” configuration information (e.g., in a “FarEndConfiguration” field as defined in section 8.3.8 of the Common Management Interface Specification, revision 5.0 published May 8, 2021), and/or other information that would be apparent to one of skill in the art in possession of the present disclosure. However, while a specific identifier has been described, one of skill in the art in possession of the present disclosure will appreciate how other identifiers will fall within the scope of the present disclosure as well.

[0035] In the specific example illustrated in FIG. **4**A, the cable connector **400***a* on the breakout cable **400** is connected to the physical port **206***a* on the switch device **202**, the cable connector **400***b* on the breakout cable **400** is connected to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, and the cable connector **400***c* on the breakout cable **400** is connected to the physical port **214***b* on the networking subsystem **210** of the endpoint device **208** in order to “break out” the connection of the physical port **206***a* on the switch device **206***a* to the pair of physical ports **214***a* and **214***b* on the networking subsystem **210** of the endpoint device **208**. As such, with reference to FIG. **4**A, in an embodiment of block **302**, the port configuration engine **212** may detect the connection of the cable connectors **400***b* and **400***c* to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208** using any of a variety of automated connection detection techniques that would be apparent to one of skill in the art in possession of the present disclosure. However, as discussed above, while the automated embodiment of block **302** of the method **300** may include the port configuration engine **212** automatically detecting the connection of the cable connectors **400***b* and **400***c* to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208** without human intervention, block **302** of a manual embodiment of the method **300** may simply involve the network administrator or other user providing that connection as described above.

[0036] In another example, with reference to FIG. **5**A, the networked system **200** may include a switch device **500** that may be substantially similar to the switch device **202** described above and that may include a physical port **502** that may be substantially similar to any of the physical ports **206***a*-**206***b* described above, and a network administrator or other user may provide a cable **504** including a plurality of cable connectors **504***a* and **504***b*, as well as a cable **506** including a plurality of cable connectors **506***a* and **506***b*, to connect the networking subsystem **210** on the endpoint device **208** to the switch devices **202** and **500**, and one of skill in the art in possession of the present disclosure will appreciate how each of the cables **504** and **506** may be provided by a DAC or other “passive” cable known in the art.

[0037] In the specific examples provided herein, the cable connector **504***a* on the cable **504** may be provided by a QSFP-DD cable connector that is configured to connect to the QSFP-DD form-factor physical port **206***a*, while the cable connector **504***b* may be provided by a QSFP56 cable connector that is configured to connect to the QSFP112 form-factor physical port **214***a*, although one of skill in the art in possession of the present disclosure will appreciate how other types of cable connectors configured to connect to other types of physical ports will fall within the scope of the present disclosure as well.

[0038] Furthermore, and as described in further detail below, the cable connectors **504***a* and **504***b* may store an identifier. For example, a respective memory subsystem (e.g., an EEPROM) in each of the cable connectors **504***a* and **504***b* may be programmed with, or may otherwise store, a cable serial number of the cable **504**, along with “far end” configuration information and/or other information that would be apparent to one of skill in the art in possession of the present disclosure. Similarly, the cable connector **506***a* on the cable **506** may be provided by a QSFP-DD cable connector that is configured to connect to the QSFP-DD form-factor physical port **502**, while the cable connector **506***b* may be provided by a QSFP56 cable connector that is configured to connect to the QSFP112 form-factor physical port **214***b*, although one of skill in the art in possession of the present disclosure will appreciate how other types of cable connectors configured to connect to other types of physical ports will fall within the scope of the present disclosure as well. Furthermore, and as described in further detail below, each of the cable connectors **506***a* and **506***b* may store an identifier. For example, a respective memory subsystem (e.g., an EEPROM) in each of the cable connectors **506***a* and **506***b* may be programmed with, or may otherwise store, a cable serial number of the cable **506**, along with “far end” configuration information and/or other information that would be apparent to one of skill in the art in possession of the present disclosure. However, while specific identifiers have been described, one of skill in the art in possession of the present disclosure will appreciate how other identifiers will fall within the scope of the present disclosure as well.

[0039] In the specific example illustrated in FIG. **5**A, the cable connectors **504***a* and **504***b* on the cable **504** are connected to the physical port **206***a* on the switch device **202** and the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, respectively, and the cable connectors **506***a* and **506***b* on the cable **506** are connected to the physical port **502** on the switch device **500** and the physical port **214***b* on the networking subsystem **210** of the endpoint device **208**, respectively. As such, with reference to FIG. **5**A, in an embodiment of block **302**, the port configuration engine **212** may detect the connection of the cable connector **504***b* to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, as well as the connection of the cable connector **506***b* to the physical port **214***b* on the networking subsystem **210** of the endpoint device **208**, using any of a variety of automated connection detection techniques that would be apparent to one of skill in the art in possession of the present disclosure. However, as discussed above, while the automated embodiment of block **302** of the method **300** may include the port configuration engine **212** automatically detecting the connection of the cable connector **504***b* to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, as well as the connection of the cable connector **506***b* to the physical port **214***b* on the networking subsystem **210** of the endpoint device **208**, without human intervention, block **302** of a manual embodiment of the method **300** may simply involve the network administrator or other user providing that connection as described above.

[0040] The method **300** then proceeds to decision block **304** where it is determined whether each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable. With reference to FIG. **4**B, in an embodiment of decision block **304** and in response to detecting the connection of respective cable connectors **400***b* and **400***c* to the plurality of physical ports **214***a* and **214***b*, the port configuration engine **212** may perform identifier retrieval operations **402** that include retrieving an identifier from each of the cable connectors **400***b* and **400***c* that are connected to the physical ports **214***a* and **214***b*, respectively, (e.g., via a respective side-band I2C bus as described above) and determining whether those cable connectors **400***b* and **400***c* are included on the same cable **400** by determining whether those identifiers match each other. As described above, each of the cable connectors **400***a*, **400***b*, and **400***c* on the cable **400** may store a common identifier (e.g., a cable serial number of the cable **400**) in a respective memory subsystem (e.g., an EEPROM) in that cable connector, and thus this example of decision block **304** provides an embodiment in which the port configuration engine **212** will determine that the cable connectors **400***b* and **400***c* are included on the same cable **400** by determining that the identifiers retrieved from those cable connectors are the same (i.e., they each identify the cable serial number of the cable **400**).

[0041] Furthermore, as described above, each of the cable connectors **400***a*, **400***b*, and **400***c* on the cable **400** may store “far end” configuration information, and at decision block **304** the port configuration engine **212** retrieve that “far end” configuration information along with the identifiers. As will be appreciated by one of skill in the art in possession of the present disclosure, the use of the “far end” configuration information along with the identifiers allows for the identification of a “break out” cable as opposed to a “point-to-point” cable that may be looped back to the same networking subsystem (and thus provides cable connectors with the same identifiers connected to the physical ports **214***a* and **214***b*, respectively).

[0042] With reference to FIG. **5**B, in another embodiment of decision block **304** and in response to detecting the connection of respective cable connectors **504***b* and **506***b* to the plurality of physical ports **214***a* and **214***b*, the port configuration engine **212** may perform identifier retrieval operations **508** that include retrieving an identifier from each of the cable connectors **504***b* and **506***b* that are connected to the physical ports **214***a* and **214***b*, respectively, (e.g., via a respective side-band I2C bus as described above) and determining whether those cable connectors **504***b* and **506***b* are included on the same cable by determining whether those identifiers match each other. As described above, each of the cable connector **504***b* on the cable **504** and the cable **506***b* on the cable **506** may store a respective identifier (e.g., a respective cable serial number) of the cables **504** and

506, respectively, in a respective memory subsystem (e.g., an EEPROM) in that cable connector, and thus this example of decision block **304** provides an embodiment in which the port configuration engine **212** will determine that the cable connectors **504***b* and **506***b* are not included on the same cable by determining that the identifiers retrieved from those cable connectors are not the same (i.e., they identify the cable serial numbers of the cables **504** and **506**, respectively).

[0043] As described above, the automated embodiment of decision block **304** of the method **300** may include the port configuration engine **212** automatically determining whether the cable connectors that were connected to the physical ports on the networking subsystem are included on the same cable, and may be performed in response to detecting the connection of those cable connectors to the physical ports **214***a* and **214***b* and without human intervention. However, decision block **304** of a manual embodiment of the method **300** may simply involve the network administrator or other user recognizing whether they have connected cable connectors on the same cable or different cables to the physical ports **214***a* and **214***b* on the networking subsystem **210** of the endpoint device **208**. Furthermore, as described below, some embodiments may provide for the automated determination that cable connectors that were connected to the physical ports on the networking subsystem are included on the same cable similarly as described above, with that automated determination resulting in the provisioning of an option for manual configuration of the connection between the networking subsystem and the switch device.

[0044] If, at decision block **304**, it is determined that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, the method **300** proceeds to block **306** where the networking subsystem configures a single networking logical port to transmit data between each of the plurality of networking physical ports and a switch physical port on a switch device. With reference to FIG. **4**C, in an embodiment block **306** and following the determination that the cable connectors **400***b* and **400***c* are included on the same cable **400** as described above with reference to the embodiment illustrated in FIG. **4**B, the port configuration engine **212** may perform logical port provisioning operations **404** that, in the illustrated embodiment, includes creating a logical port **406** that is “coupled” to each of the physical ports **214***a* and **214***b* (each of the four lanes of the physical port **214***a* and the four lanes of the physical port **214***b*), and configuring that logical port **406** to transmit data between each of the physical ports **214***a* and **214***b* and the physical port **206***a* on the switch device **202**. As such, one of skill in the art in possession of the present disclosure will appreciate how the logical port provisioning operations **404** may include logical port creation operations such as mapping all associated physical port lanes (e.g., as determined by the port configuration engine **404** for the physical ports **214***a* and **214***b*) to the logical port **406**, [logical port configuration operations such as configuring the port bandwidth and port type of the logical port **406** to match the aggregate of the physical lane speed of the physical ports **214***a* and **214***b* and describing the logical port **406** to software running on the endpoint device **208**, and/or any of a variety of other logical port provisioning operations and/or logical port configuration operations that one of skill in the art in possession of the present disclosure would recognize as providing the logical port functionality described below.

[0045] As described above, the automated embodiment of block **306** of the method **300** may include the port configuration engine **212** automatically configuring the logical port **406**, and may be performed in response to determining that the cable connectors **402***b* and **402***c* connected to the physical ports **214***a* and **214***b*, respectively, are included on the same cable **400** and without human intervention (e.g., with the need for a network administrator or other user to run a script to perform any manual software link aggregation configuration operations and/or perform any other manual configuration operations known in the art). However, block **306** of a manual embodiment of the method **300** may simply involve the network administrator or other user manually creating and configuring the logical port **406** after connecting the cable connectors **400***b* and **400***c* on the same cable **400** to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208**. Furthermore, as discussed above, in some manual embodiments the connection of the cable connectors **400***b* and **400***c* on the same cable **400** to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208** may be automatically determined similarly as described above, and may operate to expose a "400G mode" (i.e., an option to utilize all 400G of the effective data transmission rate capability of the physical port **206***a* on the switch device **202**) in a list of manual settings that are accessible to the network administrator or other user, allowing for the selection of that 400G mode to manually provide the logical port **406**.

[0046] As such, one of skill in the art in possession of the present disclosure will appreciate how, following block **306**, data may be transmitted between the physical port **206***a* on the switch device **202** and the logical port **406** via the physical ports **214***a* and **214***b* on the networking subsystem **210** of the endpoint device **208**. To continue with the specific example provided above in which the physical port **206***a* is a QSFP-DD form factor physical port that includes eight 56 Gbps lanes for an effective data transmission rate capability of 400G, and each of the physical ports **214***a* and **214***b* are QSFP112 form factor physical ports that include four 112 Gbps lanes for an effective data transmission rate capability of 400G, one of skill in the art in possession of the present disclosure will appreciate that the logical port **406** may operate to support an effective data transmission rate of 50G per lane for each of the four lanes of the physical port **214***a* and each of the four lanes of the physical port **214***b*, thus utilizing all of the 400G effective data transmission rate capability of the physical port **206***a* on the switch device **202**. Thus, FIG. **4**C provides an example of the configuration of the connection between the "first generation" switch device **202** and the "second generation" networking subsystem **210** in the endpoint device **208**.

[0047] Furthermore, while the specific example above describes the connection of an eight lane switch physical port providing an effective data transmission rate capability of 400G to a pair of 4 lane networking physical ports providing an effective data transmission rate of 200G each, one of skill in the art in possession of the present disclosure will appreciate how the teachings of the present disclosure may be utilized with switch physical ports and networking physical ports having different numbers of lanes and providing different effective data transmission rate capabilities while remaining within the scope of the present disclosure as well.

[0048] If, at decision block **306**, it is determined that each of the respective cable connectors connected to the plurality of networking physical ports are not included on the same cable, the method **300** proceeds to block **308** where the networking subsystem configures a respective networking logical port to transmit data between each of the plurality of networking physical ports and respective switch physical ports on respective switch devices. With reference to FIG. **5**C, in an embodiment block **308** and following the determination that the cable connectors **504***b* and **506***c* are not included on the same cable as described above with reference to the embodiment illustrated in FIG. **5**B, the port configuration engine **212** may perform logical port provisioning operations **510** that, in the illustrated embodiment, includes creating a logical port **512** that is "coupled" to the physical port **214***a* (each of the four lanes of the physical port **214***a*) and a logical port **514** that is "coupled" to the physical port **214***b* (each of the four lanes of the physical port **214***b*), configuring the logical port **512** to transmit data between the physical port **214***a* and the physical port **206***a* on the switch device **202**, and configuring the logical port **514** to transmit data between the physical port **214***b* and the physical port **502** on the switch device **500**. As such, one of skill in the art in possession of the present disclosure will appreciate how the logical port provisioning operations **510** may include logical port creation operations such as assigning each associated physical port lane (e.g., as determined by the port configuration engine **404** for the physical ports **214***a* and **214***b*) to the logical ports **512** and **514**, respectively, logical port configuration operations such as configuring the port bandwidth and port type of the logical ports **512** and **514** to match the aggregate of the physical lane speed of the physical ports **214***a* and **214***b*, respectively, and describing the logical ports **512** and **514** to software running on the endpoint device **208**, and/or any of a variety of other logical port provisioning operations/or logical port configuration operations that one of skill in the art in possession of the present disclosure would recognize as providing the logical port functionality described below.

[0049] As described above, the automated embodiment of block **306** of the method **300** may include the port configuration engine **212** automatically configuring the logical ports **512** and **514**, and may be performed in response to determining that the cable connectors **504***b* and **506***b* connected to the physical ports **214***a* and **214***b*, respectively, are not included on the same cable and without human intervention. However, block **306** of a manual embodiment of the method **300** may simply involve the network administrator or other user manually creating and configuring the logical ports **512** and **514** after connecting the cable connectors **504***b* and **506***b* on different cables **504** and **506**, respectively, to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208**.

[0050] Furthermore, similarly as discussed above, in some manual embodiments the connection of the cable connectors **504***b* and **506***b* on the different cables **504** and **506** to the physical ports **214***a* and **214***b*, respectively, on the networking subsystem **210** of the endpoint device **208** may be automatically determined similarly as described above, and may operate to expose a "200G mode" (i.e., an option to utilize 200G of the effective data transmission rate capability of each of the physical port **206***a* on the switch device **202** and the physical port **502** on the switch device **500**) in a list of manual settings that are accessible to the network administrator or other user, allowing for the selection of that 200G mode to manually provide the logical ports **512** and **514**. Further still, in the event a user attempts to manually configure the physical ports **214***a* and **214***b* configured as illustrated in FIGS. **5**A-**5**C with a single logical port (as described above with reference to FIGS. **4**A-**4**C), the port configuration engine **212** may display a warning that such a configuration is invalid or otherwise not recommended.

[0051] As such, one of skill in the art in possession of the present disclosure will appreciate how, following block **308**, data may be transmitted between the physical port **206***a* on the switch device **202** and the logical port **512** via the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, as well as between the physical port **502** on the switch device **500** and the logical port **514** via the physical port **214***b* on the networking subsystem **210** of the endpoint device **208**. To continue with the specific example provided above in which the physical port **206***a* is a QSFP-DD form factor physical port that includes eight 56 Gbps lanes for an effective data transmission rate capability of 400G, and the physical port **214***a* is a QSFP112 form factor physical port that includes four 112 Gbps lanes for an effective data transmission rate capability of 400G, one of skill in the art in possession of the present disclosure will appreciate that the logical port **512** may operate to support an effective data transmission rate of 50G per lane for each of the four lanes of the physical port **214***a*, thus utilizing 200G of the 400G effective data transmission rate capability of the physical port **206***a* on the switch device **202** (i.e., while stranding 200G of that 400G effective data transmission rate capability of the physical port **206***a* on the switch device **202**).

[0052] Similarly, in the event the physical port **502** is a QSFP-DD form factor physical port that includes eight 56 Gbps lanes for an effective data transmission rate capability of 400G (i.e., the physical port **502** on the switch device **500** is the same as the physical ports **206***a* and **206***b* on the switch device **202**), with the physical port **214***a* being a QSFP112 form factor physical port that includes four 112 Gbps lanes for an effective data transmission rate capability of 400G, one of skill in the art in possession of the present disclosure will appreciate that the logical port **514** may operate to support an effective data transmission rate of 50G per lane for each of the four lanes of the physical port **214***b*, thus utilizing 200G of the 400G effective data transmission rate capability of the physical port **502** on the switch device **500** (i.e., while stranding 200G of that 400G effective data transmission rate capability of the physical port **502** on the switch device **500**).

[0053] As discussed above, the two configurations illustrated and described above with reference to

FIGS. **4**A-**4**C and FIGS. **5**A-**5**C are just a few of the networking subsystem/switch device connections configurations that may be provided between the switch device **202** and the endpoint device **208** in the networking subsystem/switch device connection configuration system of the present disclosure. For example, with reference to FIG. **6**A, the networked system **200** may include an endpoint device **600** that may be substantially similar to the endpoint device **208** described above and that may include a physical port **602** that may be substantially similar to any of the physical ports **214***a*-**214***b* described above, and a network administrator or other user may provide a breakout cable **604** including a plurality of cable connectors **604***a*, **604***b*, and **604***c* to connect the switch device **202** to the endpoint device **208** and the endpoint device **600**, and one of skill in the art in possession of the present disclosure will appreciate how the breakout cable **604** may be provided by a DAC or other "passive" cable known in the art.

[0054] In the specific examples provided herein, the cable connector **604***a* on the breakout cable **604** may be provided by a QSFP-DD cable connector that is configured to connect to the QSFP-DD form-factor physical port **206***a*, the cable connector **604***b* may be provided by a QSFP56 cable connector that is configured to connect to the QSFP112 form-factor physical port **214***a*, and the cable connector **604***c* may be provided by a QSFP56 cable connector that is configured to connect to the QSFP112 form-factor physical port **602**, although one of skill in the art in possession of the present disclosure will appreciate how other types of cable connectors configured to connect to other types of physical ports will fall within the scope of the present disclosure as well. Furthermore, and as described in further detail below, the cable connectors **604***a*, **604***b*, and **604***c* may store an identifier. For example, a respective memory subsystem (e.g., an EEPROM) in each of the cable connectors **604***a*, **604***b*, and **604***c* may be programmed with, or may otherwise store, a cable serial number of the breakout cable **604**.

[0055] In the specific example illustrated in FIG. **6**A, the cable connectors **604***a*, **604***b*, and **604***c* on the breakout cable **604** are connected to the physical port **206***a* on the switch device **202**, the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**, and the physical port **602** on the endpoint device **600**, respectively. As such, with reference to FIG. **6**A, in an embodiment of block **302**, the port configuration engine **212** may automatically detect the connection of the cable connector **604***b* to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**. However, as discussed above, while the automated embodiment of block **302** of the method **300** may include the port configuration engine **212** detecting the connection of the cable connector **604***b* to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208** without human intervention, block **302** of a manual embodiment of the method **300** may simply involve the network administrator or other user providing that connection as described above.

[0056] With reference to FIG. **6**B, in response to detecting the connection of the cable connector **604***b* to the physical port **214***a* (e.g., without the connection of any cable connectors to the physical port **214***b* in this example), the port configuration engine **212** may perform logical port provisioning operations **608** that, in the illustrated embodiment, includes creating a logical port **610** that is "coupled" to the physical port **214***a* (each of the four lanes of the physical port **214***a*), and configuring the logical port **610** to transmit data between the physical port **214***a* and the physical port **206***a* on the switch device **202**. As such, one of skill in the art in possession of the present disclosure will appreciate how the logical port provisioning operations **608** may include logical port creation operations such as assigning the associated physical port lane (e.g., as determined by the port configuration engine **404** for the physical port **214***a*) to the logical port **610**, logical port configuration operations such as configuring the port bandwidth and port type of the logical port **610** to match the aggregate of the physical lane speed of the physical port **214***a*, and describing the logical port **610** to software running on the endpoint device **208**, and/or any of a variety of other logical port provisioning operations that one of skill in the art in possession of the present disclosure would recognize as providing the logical port functionality described below.

[0057] Similarly as described above, an automated embodiment may include the port configuration engine **212** automatically configuring the logical port **610**, and may be performed in response to determining that the cable connector **604***b* is connected to the physical port **214***a* without a cable connector connected to the physical port **214***b* and without human intervention. However, a manual embodiment may simply involve the network administrator or other user manually creating and configuring the logical port **610** after connecting the cable connector **604***b* to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208** without connecting a cable connector to the physical port **214***b* on the networking subsystem **210** of the endpoint device **208**. Furthermore, similarly as discussed above, in some manual embodiments the connection of the cable connector **604***b* on breakout cable **604** to the physical port **214***a* on the networking subsystem **210** of the endpoint device **208** without any cable connector connected to its physical port **214***b* may be automatically determined similarly as described above, and may operate to expose a "200G mode" (i.e., an option to utilize 200G of the effective data transmission rate capability of the physical port **206***a* on the switch device **202**) in a list of manual settings that are accessible to the network administrator or other user, allowing for the selection of that 200G mode to manually provide the logical port **610**.

[0058] As such, one of skill in the art in possession of the present disclosure will appreciate how data may be transmitted between the physical port **206***a* on the switch device **202** and the logical port **610** via the physical port **214***a* on the networking subsystem **210** of the endpoint device **208**. To continue with the specific example provided above in which the physical port **206***a* is a QSFP-DD form factor physical port that includes eight 56 Gbps lanes for an effective data transmission rate capability of 400G, and the physical port **214***a* is a QSFP112 form factor physical port that includes four 112 Gbps lanes for an effective data transmission rate capability of 400G, one of skill in the art in possession of the present disclosure will appreciate that the logical port **610** may operate to support an effective data transmission rate of 50G per lane for each of the four lanes of the physical port **214***a*, thus utilizing 200G of the 400G effective data transmission rate capability of the physical port **206***a* on the switch device **202** (i.e., while leaving 200G of that 400G effective data transmission rate capability of the physical port **206***a* available for the endpoint device **600**).

[0059] Thus, systems and methods have been described that provide for the configuration of a connection between a relatively newer generation networking subsystem via its networking physical ports and a relatively older generation switch device via its switch physical ports that have more lanes and support a lower data transmission rate than those networking physical ports. For example, the networking subsystem/switch device connection configuration system of the present disclosure may include a switch device having a switch physical port, and an endpoint device including a networking subsystem having a plurality of networking physical ports. The networking subsystem detects connection of respective cable connectors to each of the plurality of networking physical ports. If the networking subsystem determines that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, it configures a single networking logical port to transmit data between each of the plurality of networking physical ports and the switch physical port. As such, older generation switch devices having switch physical ports with more lanes supporting lower data transmission rates may be connected via a passive cable to newer generation networking devices having networking physical ports with fewer lanes supporting higher data transmission rates, with some embodiments auto-provisioning such connection configurations to eliminate the need for manual configurations that would complicate performing such connection configurations at scale.

[0060] Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

Claims

1. A networking subsystem/switch device connection configuration system, comprising: a switch device that includes a plurality of switch physical ports; an endpoint device; and a networking subsystem that is included in the endpoint device, that includes a plurality of networking physical ports, and that is configured to: detect connection of respective cable connectors to each of the plurality of networking physical ports; determine that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable; and configure, in response to determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, a single networking logical port to transmit data between each of the plurality of networking physical ports and a first switch physical port that is included in the plurality of switch physical ports.
2. The system of claim 1, wherein the first switch physical port includes eight lanes that each provide a first data transmission rate, and wherein the plurality of networking physical ports include a pair of networking physical ports each having four lanes that each provide a second data transmission rate that is greater than the first data transmission rate.
3. The system of claim 2, wherein the first switch physical port is a Quad Form-Factor Pluggable Double Density (QSFP-DD) form-factor physical port, and wherein the networking physical port is a Quad Form-Factor Pluggable 112 Gbps (QSFP112) form-factor physical port.
4. The system of claim 1, wherein the cable is a Direct Attach Cable (DAC), and wherein each of the respective cable connectors are DAC connectors.
5. The system of claim 1, wherein the cable is a break-out cable that connects the switch physical port to each of the plurality of networking physical ports.
6. The system of claim 1, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable includes: retrieving, from each of the respective cable connectors connected to the plurality of networking physical ports, a respective cable identifier; and determining that each of the respective cable identifiers is the same.
7. The system of claim 1, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable and, in response, configuring the single networking logical port to transmit data between each of the plurality of networking physical ports and the first switch physical port is performed automatically in response to detecting the connection of the respective cable connectors to each of the plurality of networking physical ports and without human intervention.
8. An Information Handling System (IHS), comprising: a processing system; and a memory system that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a port configuration engine that is configured to: detect connection of respective cable connectors to each of a plurality of networking physical ports that are coupled to the processing system; determine that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable; and configure, in response to determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, a single networking logical port to transmit data between each of the plurality of networking physical ports and a switch physical port on a switch device.
9. The IHS of claim 7, wherein the switch physical port includes eight lanes that each provide a first data transmission rate, and wherein the plurality of networking physical ports include a pair of networking physical ports each having four lanes that each provide a second data transmission rate that is greater than the first data transmission rate.
10. The IHS of claim 9, wherein the switch physical port is a Quad Form-Factor Pluggable Double

Density (QSFP-DD) form-factor physical port, and wherein the networking physical port is a Quad Form-Factor Pluggable 112 Gbps (QSFP112) form-factor physical port.

11. The IHS of claim 7, wherein the cable is a Direct Attach Cable (DAC), and wherein each of the respective cable connectors are DAC connectors.

12. The IHS of claim 7, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable includes: retrieving, from each of the respective cable connectors connected to the plurality of networking physical ports, a respective cable identifier; and determining that each of the respective cable identifiers is the same.

13. The IHS of claim 7, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable and, in response, configuring the single networking logical port to transmit data between each of the plurality of networking physical ports and the switch physical port is performed automatically in response to detecting the connection of the respective cable connectors to each of the plurality of networking physical ports and without human intervention.

14. A method for configuring a connection between a networking subsystem and a switch device, comprising: detecting, by a networking subsystem, connection of respective cable connectors to each of a plurality of networking physical ports that are included on the networking subsystem; determining, by the networking subsystem, that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable; and configuring, by the networking subsystem in response to determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable, a single networking logical port to transmit data between each of the plurality of networking physical ports and a switch physical port on a switch device.

15. The method of claim 14, wherein the switch physical port includes eight lanes that each provide a first data transmission rate, and wherein the plurality of networking physical ports include a pair of networking physical ports each having four lanes that each provide a second data transmission rate that is greater than the first data transmission rate.

16. The method of claim 15, wherein the switch physical port is a Quad Form-Factor Pluggable Double Density (QSFP-DD) form-factor physical port, and wherein the networking physical port is a Quad Form-Factor Pluggable 112 Gbps (QSFP112) form-factor physical port.

17. The method of claim 14, wherein the cable is a Direct Attach Cable (DAC), and wherein each of the respective cable connectors are DAC connectors.

18. The method of claim 14, wherein the cable is a break-out cable that connects the switch physical port to each of the plurality of networking physical ports.

19. The method of claim 14, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable includes: retrieving, by the networking subsystem from each of the respective cable connectors connected to the plurality of networking physical ports, a respective cable identifier; and determining, by the networking subsystem, that each of the respective cable identifiers is the same.

20. The method of claim 14, wherein the determining that each of the respective cable connectors connected to the plurality of networking physical ports are included on the same cable and, in response, configuring the single networking logical port to transmit data between each of the plurality of networking physical ports and the switch physical port is performed automatically in response to detecting the connection of the respective cable connectors to each of the plurality of networking physical ports and without human intervention.